

Introduction

Baroid Core Homework 9

Assessment Feedback

Congratulations, you passed.

Total score: 108 out of 111, 97%

Question Feedback

Question

1 of 46

Which of the following are characteristics of montmorillonite clay?

- ☒ Commonly used in fresh water systems
- ☒ Used for viscosity
- ☐ Form gel structures in salt water
- ☒ Produces shear thinning fluid properties

3 out of 3

Question

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What are the benefits to using a sodium-based montmorillonite clay over a calcium-based montmorillonite clay in your drilling fluid system?

- ☒ High hydration potential
- ☒ Encourages layer separation
- ☐ Low hydration potential
- ☐ Inhibits hydration potential

2 out of 2

Question

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Select the factors that affect polymer performance:

- ☐ Biological attack
- ☒ Electrolytes in solution
- ☐ Polymer color
- ☒ Polymer molecular weight
- ☐ Turbidity of solution
- ☒ Polymer charge
- ☒ pH of water

4 out of 4

Question

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Natural organic polymers ▼ are proteins, carbohydrates and untreated cellulose that can be obtained from nature.

1 out of 1

Question

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Modified organic polymers ▼ are usually natural polymers that have been chemically altered to improve specific properties.

1 out of 1

Question

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Synthetic polymers ▼ are built from inorganic compounds including petrochemicals.

1 out of 1

Question

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What are some of the uses of the polymers added to drilling fluids?

- ☒ Decrease viscosity
- ☐ Destabilize formations
- ☒ Increase viscosity
- ☐ Increase the amount of filtrate

2 out of 2

Question

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Which pH range gives excellent clay hydration?

- ☐ 6.5 – 8.5
- ☒ 9.5 – 10.5
- ☐ 12.5 – 14
- ☐ 2 – 4

1 out of 1

Question

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Adding thinners to a flocculated drilling fluid result in the clays becoming?

- ☒ Deflocculated
- ☐ Hydrated
- ☐ Flocculated
- ☐ Aggregated

1 out of 1

Question

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The clay state of edge-to-edge or edge-to-face association of clay particles is called?

- ☐ Aggregated
- ☐ Semi-dispersed
- ☐ Hydrated
- ☐ Dispersed
- ☒ Flocculated
- ☐ Deflocculated

1 out of 1

Question

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What clay state is present after the break up of the clay platelets, which results in a greater number of particles in the fluid, with an increase in surface area, and a rise of the plastic viscosity?

- ☒ Dispersed
- ☐ Flocculated
- ☐ Deflocculated
- ☐ Aggregated

Question

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When clays enter in contact water and the layers of the individual clay particles swell, the condition is called? —

- ☐ Flocculation
- ☐ Deflocculation
- ☒ Hydration
- ☐ Dispersion

1 out of 1

Question

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The face-to-face stacking of layers in the clay particles, (like in the sacks) is called? —

- ☐ Dispersed
- ☐ Hydrated
- ☐ Deflocculated
- ☒ Aggregated
- ☐ Flocculated

1 out of 1

Question

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The OH^- or hydroxide ion or hydroxyl ion needs to be present for which of the following products to work? —

- ☐ Flocculants
- ☐ Surfactants
- ☒ Most Thinners (Lignosulfonates)

1 out of 1

Question

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High molecular weight polymers are used as? —

- ☐ Deflocculants
- ☒ Viscosifiers
- ☒ Flocculants

☐ Filtration Control

2 out of 2

Question

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Low molecular weight polymers are used as?

- ☒ Thinners
- ☐ Shale Stabilizers
- ☒ Deflocculants
- ☐ Flocculants

2 out of 2

Question

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Select the reasons clays are added to drilling fluids?

- ☐ It is the best viscosifier in the market
- ☒ To provide some fluid loss control
- ☒ It is the cheapest viscosifier
- ☐ To provide density to drilling fluid
- ☒ To provide viscosity

3 out of 3

Question

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Select the properties that can be affected by addition of clays to drilling fluids:

- ☒ % solids
- ☐ Alkalinity
- ☐ Oil water ratio
- ☒ Density
- ☒ Rheology
- ☒ Fluid Loss
- ☐ Total hardness

4 out of 4

Question

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When building a WBF, select which products you would use to increase the density of the system:

- ☒ BARACARB
- ☐ BARAWEIGHT
- ☐ BARAZAN D PLUS
- ☒ BAROID
- ☐ BARANEX
- ☐ BARADEFOAM

2 out of 2

Question

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When building a WBF, select which products you would use to increase the viscosity of the system:

- ☒ BARAVIS
- ☐ PAC-L
- ☐ DEXTRID
- ☐ QUIK-THIN
- ☐ THERMA-THIN
- ☒ BARAZAN D PLUS

2 out of 2

Question

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When building and treating WBF, select which products could be used to increase the pH:

- ☒ NaOH
- ☐ KCl
- ☒ KOH
- ☐ CaCl_2
- ☐ NaCl
- ☐ NaCl

2 out of 2

Question

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When treating a WBF, select which products you would use to decrease the fluid loss of the system:

- ☐ EZ-FILTER
- ☒ CARBONOX

- ☐ IMPERMUD
- ☒ DEXTRID
- ☒ PAC-R
- ☒ FILTER-CHEK

4 out of 4

Question

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When treating a WBF, select which products you would use to decrease the viscosity of the system:

- ☐ AQUAGEL GOLD SEAL
- ☐ BARO-THIN
- ☒ THERMA-THIN
- ☒ QUIK-THIN

2 out of 2

Question

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While maintaining 2900 bbls volume, increase the weight of a 12.0 ppg drilling fluid to 15.0 ppg

For Mass Balance Exercise 1, how many sacks of barite (100lb/sack) are required? 5556

3 out of 3

Question

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For Mass Balance Exercise 1, what is the volume of fluid to dump?

Volume of fluid to dump = 377.44

3 out of 3

Question

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905 bbl of 11.4 ppg drilling fluid will be weighted up to 12.9 ppg with barite.

For Mass Balance Exercise 2, how many sacks of barite (100lb/sack) are needed? = 903

3 out of 3

Question

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For Mass Balance Exercise 2, what will be the final volume of the system after increasing the density with barite?

966.29

3 out of 3

Question

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Calculate how much water and BARACARB (CaCO_3) are required to make 500 bbls of a WBF with a final density of 13.0 ppg.

The BARACARB (SG 2.7) comes in 50lb/sack

For Mass Balance Exercise 3, what is the volume of water needed in barrels = 335.94

3 out of 3

Question

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For Mass Balance Exercise 3, how many sacks (50lb/sack) of BARACARB are needed to build the system?

Sacks of Baracarb needed = 3106

3 out of 3

Question

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Weight up 2200 bbls of 16.5 ppg drilling fluid to 18.0 ppg and maintain exactly 2200 bbls.

Evaluate how much barite (100 lb/sack) will be needed and how much reduction in volume will be required?

Sacks of barite (100lb/sack) needed are = 2619

3 out of 3

Question

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For Mass Balance Exercise 4, how much of the 16.5ppg you need to dump in order to increase the density of the system?

Reduction of volume in bbl = 177.9

3 out of 3

Question

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For Mass Balance Exercise 4, how much of the original 16.5 ppg fluid are you keeping in the pits? _____

The volume of the original 16.5ppg fluid to keep in bbl = 2022.1

3 out of 3

Question

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Given the following results from a retort test, perform a complete a solids analysis.

Retort results: %Water = 79.0 % Oil = 1.0

CEC = 5 MW = 13.2 ppg

The density of oil is 7ppg (SG = 0.84) and density of water is 8.345ppg (SG = 1).

What is the ASG? (3 decimals) 3.915

3 out of 3

Question

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For Solids Analysis Exercise 1, calculate the percentage and concentration of Low gravity solids (LGS) _____

%LGS = 3.56 LGS_{ppb} = 32.44

3 out of 3

Question

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For Solids Analysis Exercise 1, calculate the percentage and concentration of High Gravity Solids (HGS) _____

%HGS = 16.44 HGS_{ppb} = 242.01

3 out of 3

Question

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For Solids Analysis Exercise 1, calculate the concentration of drill solids in ppb _____

Drill solids ppb = 7.34

0 out of 3

Drill Solids = 32.44 - (5x5) = 32.44 - 25 = 7.44ppb

Question

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Given the following results from a retort test, perform a solids analysis by calculating the requested values below.

Fluid data: Water = 7.4 ml Oil = 0.7 ml CEC = 6 MW = 11.8 ppg

The density of oil is 7ppg (SG = 0.84) and density of water is 8.345ppg (SG = 1).

What is the ASG? 3.236

3 out of 3

Question

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For Solids Analysis Exercise 2, what is the percentage of low gravity solids? _____

%LGS = 11.45

3 out of 3

Question

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For Solids Analysis Exercise 2, calculate the concentration of High Gravity Solids in ppb _____

HGS ppb = 111.14

3 out of 3

Question

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Using the information below, calculate all the volumes and circulation parameters:

Parameters	Pump 1	Pump 2
Liner	6.5"	6.5"
Stroke	12"	12"
Efficiency	96%	96%
Stroke/minute (spm)	58	57

13 3/8" surface casing set @ 5,000' MD/TVD
8 5/8" intermediate casing set @ 14,000' MD/TVD
5 1/2" liner @ 18,000' top of liner @ 13,000' MD/TVD
4 3/4" open hole to 22,000'

Drill pipe: 4.5" x 3.958"
Drill pipe: 3.5" x 2.9" x 10,350'
Drill collar: 3.5" x 1.5" x 2,000'

Pit volume: 1,800bbl

What is the pump output in bbl/stk? 0.1183

3 out of 3

Question

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For Hole Volume Exercise 1, calculate the Annular Volume in bbl

Annular Volume in bbl = 837.26

3 out of 3

Question

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For Hole Volume Exercise 1, calculate the Drill String Volume in bbl

Drill String Volume in bbl = 235.69

3 out of 3

Question

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For Hole Volume Exercise 1, calculate the total circulating Volume in bbl

Circulating volume in bbl = 2873.05

3 out of 3

Question

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For Hole Volume Exercise 1, calculate the Bottoms up strokes and time in minutes

Bottoms up strokes = 7078 Bottoms up time in min = 62

3 out of 3

Question

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For Hole Volume Exercise 1, calculate the surface to bit strokes and time in minutes

Surface to bit strokes = 1994 Surface to bit time in min = 18

3 out of 3

Question

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For Hole Volume Exercise 1, calculate the Suction to suction strokes and time in minutes

Suction to suction strokes = 24287 Suction to suction time in min= 212

1 out of 1